MBS Convexity & the Treasury Basis Nexus

Falling Rates, Rising Risks: Hedging Pressures of 4+ Cpn MBS

stwill1

The Playing Field

- Hedge funds capitalize on persistent demand among institutional investors and asset managers (AMs) to add duration synthetically, taking the short side in U.S. Treasury futures while acquiring cheapest-to-deliver Treasury bonds, thereby capturing a basis of about 20–30 bps p.a. This strategy necessitates high leverage (typically 20–40x, and often higher) to achieve target returns, exposing funds to potential risks during liquidity crises where basis widening can lead to margin calls and forced deleveraging, potentially necessitating official intervention.

The Market's Blind Spot

- The driver of the basis trade—persistent AM demand for long Treasury futures—was largely unexamined before 2022. Recent research, including the January TBAC presentation and subsequent Federal Reserve analysis, has since begun to document the specific portfolio mechanics at play.

Reaching for Duration

- U.S. mutual funds account for 40–60% of long Treasury futures positions within the CFTC's AM/Institutional category, primarily in 2s, 5s, and 10s contracts. This behavior is concentrated in a subset of funds and is strongly linked to agency MBS holdings and their duration management.
- To capture the attractive, credit-risk-free OAS of agency MBS (30–70 bps), AMs often hold them overweight to benchmarks like the US Agg. Because agency MBS have shorter durations, these funds use long Treasury futures to close the duration gap and match benchmark sensitivities.

When the Tide Went Out: MBS in the Post-QE Era

- MBS OAS movements since 2019 reflect two primary drivers: 1) massive Fed and bank-driven supply/demand imbalances, with AMs acting as a key countercyclical shock absorber, and 2) a post-2022 spike in interest rate volatility and prepayment uncertainty, which increased the unhedgeable risk premium demanded by investors.

Negative Convexity

- Linear hedging fails for MBS because duration is dynamic: rate rallies spur refinancings, causing high-coupon MBS duration to decay and dragging active portfolios below benchmark targets.
- Closing this gap forces asset managers into programmatic Treasury futures buying. This concentrated, one-sided demand pushes futures rich to cash, widening the basis and inflicting mark-to-market losses on levered hedge fund basis books.

A chasm: Index vs. AMs' duration & convexity

- While index-level MBS data (duration ∼6 years, near-zero convexity) suggests minimal re-hedging pressures, analysis of a representative IID fund reveals significant deviations—with 62% of holdings at 5%+ coupons and 75% maturing post-2052—resulting in lower duration (3.78 years) and more negative convexity (-1.6 to -2) compared to the index.
- The divergence between IID funds and the index is attributed to active portfolio management in response to market conditions: reducing exposure as OAS contracted in 2021, then nearly doubling agency mortgage holdings to 15 percentage points above benchmark when OAS surged in 2022–2023, acquiring newly issued 4+% coupon MBS with lower durations.

The dD/dr Mismatch

- dD/dr, calculated as duration squared minus convexity ($D^2 - C$), quantifies bond duration's sensitivity to interest rate changes. High-coupon MBS portfolios can approach dD/dr values of 300. This starkly contrasts with the negligible dD/dr of Treasury futures, creating a significant second-order re-hedging mismatch.

Quantifying the Re-Hedging Requirement

A direct, security-level analysis of N-PORT filings establishes the paper's core quantitative findings:

- The MBS coupon distribution for futures-using mutual funds is distinctly double-humped, with high-coupon (4%+) securities accounting for an excess 26% of their total MBS holdings. This creates a portfolio dD/dr of 94 versus the index's 38—an excess sensitivity of 56—meaning portfolio duration shortens by an additional 0.56 years relative to the benchmark over a baseline 100 bps rate decline.
- Futures-using mutual funds hold $417 bn in MBS. With an estimated dD/dr gap of 69 versus the US Agg, a 100 bps rate decline from current levels creates a DV01 shortfall of $28.8 million.
- With their Treasury futures hedges carrying a weighted duration of 4.15 years, neutralizing this $28.8 mn DV01 shortfall requires an additional $69.4 bn in net long notional for mutual funds alone.
- Including separate accounts, estimated to be of approximately equal size to the mutual fund cohort, the total asset manager DV01 shortfall rises to $57.6 mn. This requires an aggregate $139 bn in additional long Treasury futures notional to hedge a 100 bps rate decline.
- A sanity check confirms the scale of the risk: hedging the full duration loss on AMs' $260 bn of excess high-coupon MBS would require $180 bn in Treasury futures, corroborating the bottom-up model.

Impact Analysis: Stock, Flow & The March 2020 Precedent

- The impact of AMs' re-hedging pressures is contingent on timeframe and market conditions. From a stock perspective, it seems manageable, but from a flow perspective, it could significantly impact the cash-futures basis, especially if concentrated in short periods.
- March 2020 market dislocation provides context: $259 bn in basis-widening flows drove the basis to peak at 100–125 bps by March 12–13, suggesting that AMs' potential re-hedging needs of up to $200 bn could overwhelm FIRV capacity and lead to significant market disruption.

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Summary

This note identifies and quantifies a novel risk factor for the $750 bn Treasury cash-futures basis trade. The risk stems from the benchmark-driven duration re-hedging of asset managers during rate declines, a dynamic driven by their substantial overweights in recent-vintage, high-coupon agency mortgage-backed securities with deeply negative convexity.

Our analysis of N-PORT filings, supplemented by data on separate accounts with similar mandates, finds that asset managers would need to acquire an additional **$139 bn** in notional Treasury futures to neutralize the DV01 shortfall from an initial 100 bps rate decline. The maximum potential re-hedging pressure is about $180 bn.

For context, during the first two weeks of March 2020, the aggregate net basis widening pressures from all investor types totaled approximately $250 bn.

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The note first establishes the conceptual framework by reviewing the drivers of the basis trade and outlining the core convexity argument (Sections I–V). It then presents the detailed empirical analysis that quantifies the re-hedging risk (Sections VI–X).

I. The Playing Field

Q: Why are hedge funds short UST futures and why does it matter?

The rebuilding of hedge fund short positions in Treasury futures has become a focus of intense market and regulatory scrutiny over the past two years. The trade's potent mix of high-finance intrigue, historical significance, and large dollar figures involved has made it a compelling subject for both commentary and academic study.

Alongside a number of high-profile speeches[1] and policy proclamations,[2] recent years have produced a series of robust research papers in this area, looking at the issue from two angles: one, from a financial stability perspective, and the other, in terms of the role of hedge funds in absorbing a significant portion of still large U.S. government deficits. This analysis focuses exclusively on the financial stability perspective.[3]

The mechanical reasons for hedge funds' short positions are no secret and well described in the literature. The core mechanics are as follows:

- Institutional investors[4] and asset managers (AMs) have for much of the past decade shown a pronounced appetite for long duration exposure through U.S. Treasury futures. In terms of underlying net notional value, this demand has often exceeded $500 bn, and is currently closer to $1 tn.
- In even the most liquid derivatives markets, such a one-sided demand imbalance creates ripples that cause futures prices to rise above their cash-and-carry implied value over the underlying delivery basket. At some point, this will naturally attract profit-oriented counterparties who are prepared to take the other side of the futures leg.
- Driven by Dodd-Frank restrictions alongside tighter post-GFC leverage rules, banks largely withdrew from direct engagement in capital-intensive arbitrage, with former desk talent long since decamping to the buy side.
- It is ultimately a small group of PMs at about 10 multi-manager platforms, half a dozen fixed-income relative value (FIRV) hedge funds, and a similar number of macro shops that collectively take on much of the corresponding short positions to AMs' long futures exposure. They execute this by simultaneously acquiring a (near-) cheapest-to-deliver (CTD) Treasury bond.[5] Combined with a short futures position at the CME, this creates a package that isolates the fair value gap if held to expiry, thereby capturing the basis.

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[1]Gensler, G. "Fall Feelings: Treasury Markets' Efficiency and Resiliency," November 7, 2023. Remarks before SIFMA, Washington, D.C.

[2]Barr, M. S. "The 2023 U.S. Treasury Market Conference," November 16, 2023. Remarks at the Federal Reserve Bank of New York.

[3]The second becomes practically redundant with regard to relative value hedge funds when the end holders of total duration (including derivatives) are considered, instead of just cash positions.

[4]Such as pension funds, endowments, insurance companies, mutual funds, but explicitly excluding hedge funds.

[5]Or, in many cases, some off-the-run, even cheaper, Treasury bond.

- To achieve the high single-digit+ gross returns that practically all involved hedge funds are targeting, substantial leverage is required.[6] With annualized basis values between ∼20–30 bps, this implies leverage of 20–40x. Such are typical levels, although even higher ones are feasible provided cross-margining is possible, there is recourse to a broader portfolio, or Tier 1 clients are involved.

A significant advance in recent years is the availability of granular data, which allows these dynamics to be documented from multiple angles with little time lag. The primary constraint remains that the most comprehensive analysis requires non-public data, explaining why official institutions like the Fed, OFR, and Treasury have led the research effort.

Beyond their data advantage, there is another reason for the high level of official interest: the large notional amounts and leverage involved, as well as the fact that it is ultimately the sovereign's debt market that is the arena. The main risks of these types of basis trades are well-established: any dash-for-cash dynamic—such as March 2020,[7] or any other financial crisis where the need for immediate liquidity exceeds the concurrent flight to sovereign bonds—will put pressure on the basis to widen. The resulting mark-to-market losses, coupled with increased margin requirements on futures and larger repo haircuts, may prompt today's very risk-conscious hedge funds to de-risk. In the worst case, this could set off a self-reinforcing spiral that might eventually require official intervention. It is no surprise, then, that the Federal Reserve and fellow financial stability authorities are taking an interest.

All of the above is widely known and hardly surprising; after all, Treasury cash-futures arbitrage is anything but a new idea. Any reasonably designed internal stress test naturally accounts for these factors, making the outlined risks at worst known unknowns.

[6]Some hedge fund exposure may not be directly return seeking, but rather aimed at securing dealer balance sheet capacity. Such capacity, often deployed on a 'use-it-or-lose-it' principle given post-GFC leverage constraints, may require a fund to engage in low-risk, high-leverage warehousing trades so that leverage can be accessed deliberately when the need arises. As long as they are not too negatively carrying, repo and short Treasury futures fit this bill. A look at AMs' long futures positions in 2017 and 2021, when the basis was close to zero, suggests that this *natural* demand from hedge funds could be around $200 bn.

[7]Josh Younger and his team's research is the best real-time account of this episode and helped put the phenomenon on the map; a public rendition can be found in the "Revisiting the Ides of March" series on Brad Setser's blog. "Basis Trades and Treasury Market Illiquidity" by Daniel Barth and Jay Kahn is the essential official study.

II. The Market's Blind Spot

Q: Why are asset managers long?

We deliberately sidestepped a central question in the opening section: why do asset managers hold such large long positions in Treasury futures? The omission reflects a surprising gap in the public discourse between 2019 and 2022, when little research explored the root causes of this persistent demand. The curious aspect of this failure to examine root drivers is that the positive basis clearly indicates that demand has driven the phenomenon, with hedge funds merely providing a balance sheet service to correct a mispricing. By focusing exclusively on hedge funds, much of the analysis overlooked the agency of asset managers, the *causa prima* of the observed effects.

Such a shift in perspective is not merely of academic interest. For a FIRV portfolio manager, there is little that is more crucial than understanding the factors that influence counterparties' trading decisions, particularly during stress events. It goes without saying that sizing decisions are shaped by this understanding.

With the rebuilding of AMs' net long positions in 2022, a more nuanced, causal view began to emerge. One could hear it in Ken Griffin's pointed comments last November.[8] Craig Pirrong built on this in a post.[9] As the discussion gained momentum, the January TBAC presentation[10] provided an insightful, graph-rich analysis that shed light on the dynamics at play from both the AM and hedge fund sides.

The presentation outlined three potential drivers for asset managers' long futures positions, ranging from tactical market-timing to structural portfolio construction:

- Tactical positioning (outright duration or curve views).
- Operational liquidity management (duration offsets for cash held against redemptions).
- Strategic duration-matching (synthetic duration to offset lower-duration spread assets and match a benchmark).

While TBAC laid out the structural duration-matching thesis persuasively, its analysis remained anchored in aggregate market data and concluded with a call for deeper study. And indeed, the actual footprint was never entirely hidden, lying dispersed across mandatory public filings where AMs document their underlying exposures. Navigating this fragmented disclosure trail,

[8]Ken Griffin quoted in "Citadel's Ken Griffin warns against hedge fund clampdown to curb basis trade risk," *Financial Times*, November 5, 2023.

[9]Pirrong, C. "The Basis for the Treasury Basis Trade: Leverage Laundering?," *Streetwise Professor* (blog), November 7, 2023.

[10]U.S. Department of the Treasury. "Treasury Borrowing Advisory Committee: Charge 1," presentation for the meeting of January 30, 2024.

which connects broader behavioral motives directly to individual positioning, offers the most reliable way to substantiate the dynamic.

III. Reaching for Duration

Q: How does MBS exposure lead to long UST futures?

While surveying mutual funds through semi-annual reports was technically feasible, an exercise not lost on reputable FIRV desks, doing so comprehensively was a non-trivial task. The process was hindered by the challenge of locating older documents and ensuring the sample was representative of the broader universe.

A recent working paper by four Fed economists provides the necessary breakthrough.[11] This paper largely resolves the data challenges and causal questions that had been a hurdle.

To lay the groundwork for their analysis, they made a crucial decision: mostly setting aside the semi-annual, non-standardized mutual fund documents and instead opting for quarterly SEC N-PORT filings. These reports, which are publicly available through the SEC's EDGAR, provide a standardized format and a comprehensive line-by-line listing of the holdings of all U.S. domiciled mutual funds. Following the heavy lifting—as Section VIII. will later confirm—of consolidating all this data into a single database, they then linked individual bond CUSIPs to ICE pricing data to provide an accurate and up-to-date portfolio snapshot of the sector four times a year, starting in late 2019.

Summarizing an 80-page document is a challenging task. We focus here on the core empirical findings that underpin our subsequent quantitative modeling. For a more detailed understanding, consultation of the original paper is strongly encouraged. To support the discussion, we will draw heavily from the tables and figures in the back pages of the Fed working paper.[12]

1) N-PORT-filing U.S. mutual funds tend to account for between 40% and 60% (Figure 2) of the respective long and short, and hence net, Treasury futures positions recorded in the CFTC's *Traders in Financial Futures: Asset Manager/Institutional* category. The long positions are particularly concentrated in 2s, 5s, and 10s contracts. Moreover, since many of these fund managers also oversee separate accounts or international vehicles with similar mandates,[13] the conclusions drawn are all the more representative.
2) The paper lands squarely on the last option as the most plausible explanation: Funds acquire spread assets with below-benchmark durations and use long Treasury futures to adjust their overall duration to match a given benchmark. In the aggregate, this behavior proves highly reliable (Figure 3).
3) This pattern of behavior is not seen across all funds, but concentrated in a limited, yet highly time-consistent, cohort of futures-using funds. These are almost inevitably at the non-index-hugging end of the spectrum, are more total-return-focused, trade more frequently, and have higher fees than their more humdrum brethren.
4) For futures users, the principal asset driving this futures usage is agency mortgage-backed securities (MBS). As shown in Figures 1 and 4, intermediate investment grade (IID) debt funds—the largest and most futures-rich fund sub-category—typically have above-benchmark exposure to agency and private-label MBS, as well as other types of ABS. However, only agency MBS exhibits pronounced fluctuations, exceeding 10% of total assets. Exposure to these securities was high at the end of 2019, dipped below benchmark weights in 2021, and has now returned to record levels.
5) Agency MBS, as Figure 1 illustrates, typically have shorter durations than the Bloomberg US Aggregate Bond Index (the "Agg") or similar benchmarks. As a result, funds that overweight these bonds but want to maintain benchmark sensitivities must add duration elsewhere. Treasury futures provide the most efficient mechanism to close this gap, directly matching the observed behavior of IID funds and the CFTC's AM category, whose net long positions in futures mirrored the MBS allocation: rising to a peak in 2019, falling to a trough in 2022, and then surging to new highs in 2023 and 2024.

[11] Barth, D., Kahn, R. J., Monin, P., & Sokolinskiy, O. "Reaching for Duration and Leverage in the Treasury Market," (2024). Finance and Economics Discussion Series 2024-039. Washington, D.C.: Board of Governors of the Federal Reserve System.

[12] The numbering used to reference these tables and figures corresponds to the numbering in this paper, not the original Fed paper.

[13] See for instance: Gerakos, J., Linnainmaa, J. T. & Morse, A. "Asset Managers: Institutional Performance and Factor Exposures," *Journal of Finance, 76*(4), 2035-2075, April 15, 2021. On the basis of a large global consultant's database tracking $18 tn in separate accounts—the way most institutional investors delegate mandates to asset managers—in 2012, they show that U.S. equities and fixed income mandates comprised around 50% of the total. Taking this proportion and applying it to the latest annual survey by *Pensions & Investments* of consultants and their advised assets of around $48 tn in 2024 implies that $24 tn of U.S. assets are being managed through separate accounts. By comparison, the Federal Reserve's Z.1 release currently puts total U.S. mutual fund assets at $21 tn.

	Dec 2019		Jun 2021		Jun 2023	
	Share	Duration	Share	Duration	Share	Duration
Index funds						
U.S. Treasuries	42.99	6.4	41.51	6.79	45.07	6.04
Agency MBS	24.98	3.33	22.83	3.85	22.71	5.17
Private-label MBS	0.31	1.27	1.2	5.03	0.93	3.87
Corporate debt	22.21	8.23	24.61	9.02	22.39	7.4
CDO	0.04	0.42	0.32	1.57	0.43	2.01
Other ABS	0.05	0.13	0.01	0.06	0.01	0.04
Other	9.42	6.39	9.53	6.68	8.46	5.64
Total duration						
Cash only		6.03		6.62		6.08
Cash + Futures		6.03		6.62		6.08
2023 Non-futures holders						
U.S. Treasuries	19.25	8.71	18.08	8.18	23.63	9.19
Agency MBS	32.58	3.38	27.04	4.3	33.17	4.24
Private-label MBS	6.45	2.4	4.65	3.22	3.15	2.4
Corporate debt	22.76	6.64	28.44	7.86	23.09	6.94
CDO	0.68	0.09	1.46	0.14	0.87	0.29
Other ABS	6.92	1.67	5.85	2.18	8.93	2.04
Other	11.36	6.51	14.48	7.02	7.16	6.26
Total duration						
Cash only		5.35		6.05		5.83
Cash + Futures		5.50		6.17		5.83
2023 Futures holders						
U.S. Treasuries	21.62	9.3	29.71	9.41	21.52	9.3
Agency MBS	30.93	3.53	19.7	4.25	33.12	3.78
Private-label MBS	8.15	2.5	6.75	2.78	6.63	2.76
Corporate debt	21.76	5.56	25.24	7.38	22.81	5.95
CDO	2.21	0.78	1.85	0.83	1.68	1.01
Other ABS	2.28	1.7	2.83	2.04	3.24	1.86
Other	13.06	4.77	13.93	5.66	11.0	4.73
Total duration						
Cash only		5.06		6.28		5.28
Cash + Futures		6.11		6.76		6.09

Table 11: **Weighted average duration of intermediate investment-grade debt funds by investment.** This table shows the duration of cash holdings of intermediate investment-grade debt funds by asset category, along with the weight of each category in total net assets of the funds, and the total weighted-average duration of the funds. Results are split by index funds, non-futures holders and futures holders.

Fig. 1

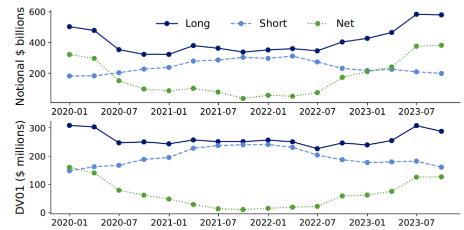

Figure 5: **Mutual fund goss and net positions in Treasury futures:** The figure shows total long, short and net Treasury futures positions of mutual funds in dollars (top panel) and DV01 (bottom panel).

Fig. 2

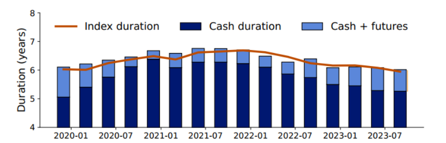

Figure 10: **Duration of cash and futures positions for intermediate investment-grade debt funds:** The figure show the difference between the cumulative change in share of different asset classes in total assets since 2015 for futures holders (bottom panel) and index funds (top panel) as of June 2023.

Fig. 3

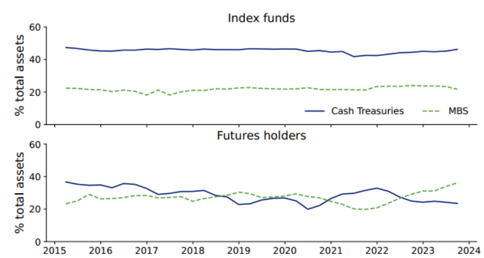

Figure 9: **Share of Treasury cash and mortgage backed securities in total assets for futures holder and index fund intermediate investment grade debt funds:** The figure show the difference between the cumulative change in share of different asset classes in total assets since 2015 for futures holders (bottom panel) and index funds (top panel) as of June 2023.

Fig. 4

6) The evident reason for benchmark deviations is the relative cheapness—and occasionally richness—of agency MBS compared to government bonds (Figure 5), as measured by an option-adjusted spread (OAS). For a fund benchmarked to a high-quality index like the Agg, the potential to capture a credit-risk-free excess spread of 30–70 bps is an attractive prospect. Achieving comparable or higher excess spread in corporate credit, meanwhile, requires venturing beyond high-grade IG into segments burdened by lower liquidity and heightened downgrade risk.

This evidence largely settles the question of causality. Active strategies—such as Core, Core Plus, and BondPLUS funds—systematically reach for yield by overweighting agency MBS, then rely on long Treasury futures to maintain benchmark duration. The data leaves little doubt that this portfolio construction practice is the primary driver of the long futures overhang.

A number of additional nuances may merit further examination, including the rationale behind funds' strict duration alignment with benchmarks, their limited engagement in repo markets, and their relatively modest use of swaps. For now, these topics will be put on hold, as more immediate concerns take priority.

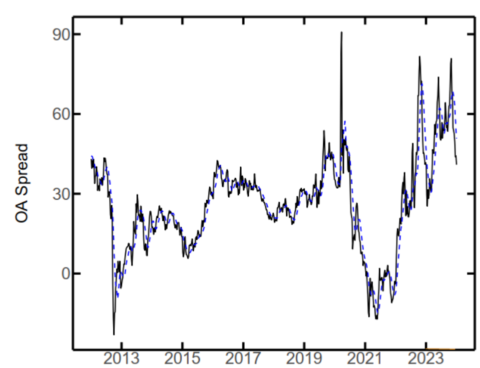

Fig. 5: The graph shows the OAS of current coupon FNMA 30y TBA, one of the most liquid mortgage instruments and therefore often preferred over cash index metrics, even if the general trends coincide. Source: Adapted from Barth et al. (2024), Figure A.13.

IV. When the Tide Went Out: MBS in the Post-QE Era

Q: MBS OAS moves. Why?

The dramatic swings in MBS Option-Adjusted Spreads since 2019 are key to understanding the variability of funds' MBS and futures positions. Examining this path through two distinct lenses—one obvious, the other less so—clarifies the underlying mechanics governing the spread's trajectory.

TABLE I: Net Issuance and Purchases of Agency MBS by Sector (in $ bn)

Sector	2020	2021	2022	2023
Net issues (a)	657	613	979	297
Federal Reserve (b)	663	576	12	−216
U.S. Banking & Credit Institutions (c)	694	606	−230	−201
Net buy/sell pressure (b)+(c)−(a)	700	570	−1,197	−714
U.S. domiciled mutual funds (d)	9	−120	42	155

Source: Financial Accounts of the United States – Z.1

The obvious lens is supply and demand, stemming from the Federal Reserve's COVID-19 response and its eventual unwinding. As Table I shows, the Fed directly acquired more than $1 tn worth of agency MBS over the course of 2020 and 2021. When combined with sovereign bond QE acquisitions, these purchases created ample liquidity, translating into excess reserves at banks. These reserves, in turn, prompted U.S. banks to acquire similar amounts of agency MBS during the same period. The cumulative effect of these two less price-sensitive demand factors was an annual purchase imbalance of over $600 bn. However, this imbalance reversed dramatically in the following two years, particularly in 2022, when large net issuance, no Fed purchases, and bank sales resulted in over $1 tn of net supply to be absorbed by others.

As the table also shows, U.S. mutual funds have acted countercyclically to Fed and bank flows, mitigating both price overshooting and, more recently, undershooting. Although the size of fund transactions is significantly smaller than that of the previous groups, it is worth noting that separate accounts and overseas funds with similar mandates are roughly equal to U.S. mutual funds in size. Together, they absorbed around 40% of the imbalance in 2021 and 2023. Notably, the years in which they failed to act as a counterbalance—2020 and 2022—saw the OAS move sharply, first downwards and then upwards.

Through this supply-and-demand lens, AMs' portfolio shifts can be attributed to central and commercial banks' actions that impacted MBS valuations. From a macro stability standpoint, the flexibility of AMs is a desirable feature, as it helps to cushion price overshooting at both extremes.

* * *

The second rationale views the OAS not as a market friction, but as a risk premium. From this perspective, the key question is what fundamental, systematic risks investors are being compensated for, and why the price of that risk has changed so dramatically.

How does this apply to MBS in recent years? Skipping early 2020 and jumping to mid-year, when the biggest market tremors had subsided, it is fairly obvious why MBS Z-spreads should compress. With rates assumed to be at the zero lower bound for a lengthy period, the value of the option component fell to levels last seen in the early 2010s. Combined with the expectation of a firm zero bound, this meant that the PnL risk of duration-hedged MBS exposures was very low, thus commanding little to no risk premium.

Then, rates started climbing, and interest rate volatility returned with a vengeance. The trickier question is why the residual spread—the OAS—increased. A properly calibrated commercial model's Option Cost internalizes the market's price of risk by calibrating to a core set of traded derivatives, removing the most readily hedgeable components of optionality. However, this process is inherently incomplete, leaving a residual of unspanned risks within the interest rate process itself—such as its higher-order dynamics and tail behavior—that the calibration can only partially capture.

The other, more prominent source of unhedgeable risk is prepayment model uncertainty, i.e. the inherent imprecision of any model that attempts to predict homeowner refinancing behavior.

What makes these distinct sources of model risk systematic is how they interact with the instrument's structure. Both lead to imperfect hedges, creating residual PnL risk. For a negatively convex instrument like an MBS, this residual risk is not benign noise; the convexity creates a structurally asymmetric payoff profile for the hedged portfolio. The potential dollar losses are inherently larger than the potential gains,[14] and these net losses are most likely to materialize during periods of market stress—precisely when the marginal value of capital is highest.

Through this risk-premium lens, the dramatic widening of OAS after mid-2022 can be seen as the market pricing a much higher premium for this dual-faceted risk, precisely because it was amplified by one of the most volatile interest rate environments in decades. Whether

[14] From a quantitative perspective, both unspanned rate dynamics and prepayment uncertainty render the true hedge ratios of the security stochastic. This imperfect replication leads to a non-zero realized quadratic variation in the PnL of the hedged portfolio, which requires a risk premium.

this justifies movements of more than 50 bps is another question, where doubts are warranted.

Regardless of the precise attribution between these factors, their combination established a clearing price for the asset's risk. This resulting spread proved wide enough to attract the new marginal buyer: mutual funds, whose mandates often permit them to hold the underlying risk directly, rather than as a pure duration-hedged arbitrage.

V. Negative Convexity

Q: How does a portfolio management tactic become a systemic vulnerability?

Thus far, this analysis has reviewed the causal factors, identified in recent Federal Reserve research, that drive Treasury cash-futures basis moves and the corresponding rise in offsetting futures positions. It has been established that AMs tend to increase their long positions in Treasury futures when they deviate from benchmark weights in MBS or, to a lesser extent, other lower-duration assets. Such shifts can be amplified when allocators shift their portfolios towards funds with mandates that allow for benchmark deviations, often resulting in a structural overweight in lower-duration assets. Both mechanisms have been at play over the past decade, offering a credible explanation for observed market developments.

In a static world where asset durations are stable, the resulting **Hedging Demand** follows a simple, linear logic:

$$\underbrace{(\text{Fund AUM} \times \text{MBS Share})}_{\text{Position Size}} \times \underbrace{(\text{Constant Duration Gap})}_{\text{Static Risk}}$$

The central thesis of our analysis, however, is that for agency MBS, this static description is critically incomplete. The defining characteristic of MBS—an uncertain prepayment profile—makes its duration highly volatile. This transforms the hedging calculation from a static problem into a dynamic one, driven by a single, potent mechanism:

> Interest rate fluctuations and the resulting prepayment behavior alter the duration profile of MBS, forcing duration-benchmarked asset managers into hedging adjustments that are, to a first order, independent of their market views.[15] These near-automatic actions, likely to be executed quickly and in significant size, create a hidden and potentially destabilizing feedback loop in the Treasury market.

This dynamic forces a critical modification to our static model, replacing the constant risk factor with a volatile one:

$$\underbrace{(\text{Fund AUM} \times \text{MBS Share})}_{\text{Position Size}} \times \underbrace{f(\text{MBS Duration})}_{\text{Dynamic Duration Risk}}$$

MBS Duration is the crucial dynamic variable. Its volatility is the engine of the re-hedging requirement, directly controlling the **Dynamic Duration Risk**. While the components of **Position Size**—Fund AUM and MBS Share—are the result of deliberate, longer-term allocation decisions, MBS Duration is not. It is the multiplication of this volatile, market-driven risk by the static, structurally-embedded position size that transforms a portfolio management tactic into a hidden systemic vulnerability.

* * *

What are the implications for the stability of basis trades, where hedge funds provide the offsetting short leg to AMs' long futures positions? Let's step through the cases.

If interest rates rise, the incentive to prepay mortgages decreases, and borrowers will prepay more slowly than expected. This extends MBS durations, allowing AMs to reduce their current long Treasury futures exposure, which tightens the basis and nets hedge funds a mark-to-market (MTM) profit. No stability issues there.

Conversely, when interest rates decline, prepayments accelerate, shortening MBS duration and prompting AMs to increase their long Treasury futures exposures. This puts widening pressure on the basis, resulting in a MTM loss for hedge funds. Decidedly the more concerning case, and worth a closer look.

What exactly would happen in reality depends largely on three key factors: (1) the magnitude of interest rate moves, (2) the amount of futures-hedged MBS, and (3) their convexity profile. Other secondary influences include the ability of hedge funds to meet such additional hedging needs, their behavior in the event of MTM losses, the willingness of other institutions to take on the short futures leg, the level at which asset managers become price sensitive enough to accept temporary

[15] While one might argue that a seasoned portfolio manager would use discretion in a crisis, this view mistakes the agent for the system. The benchmark-relative nature of institutional asset management creates a powerful institutional imperative that compels action. This system operates across three domains: (1) external client mandates that define benchmark adherence; (2) internal risk systems that enforce tracking-error constraints; and (3) career incentives that align compensation and reputation with minimizing tracking error.

duration deviations, and, as a last resort, potential Fed involvement.

These are all complex questions that can consume many hours of analysis. The next sections will focus on the first three, yet some important observations can be made upfront:

- Basis moves under stress can be driven not only by the cash leg, in which much of the March 2020 analysis is rooted, but also by the futures leg through MBS re-hedging.
- This risk is particularly pronounced during financial shocks and recessions, which dampen growth and inflation expectations, thus sending rates lower.
- Moreover, the re-hedging dynamic has near-automatic qualities, making it potentially more dangerous than well-analysed risks from the cash side, as it does not require any party to be explicitly wrong-footed.

VI.
A Chasm: Index vs. AMs' Duration & Convexity

Q: Why do AMs not hold index-like MBS?

To determine whether this novel risk channel is a purely theoretical one or indeed of real-world significance, answers are needed to the three questions from earlier:

- the magnitude of rate swings
- the amount of MBS hedged with Treasury futures
- the convexity of AMs' MBS holdings

Rough priors exist for the first two. Rates can trivially fall tens of, and in some historical cases hundreds, of basis points in a matter of weeks in stress scenarios. AMs' MBS holdings, as the Fed paper showed, are substantial, amounting to hundreds of billions of USD, even if the precise amount, especially of above-benchmark holdings is not yet known. None of these risks materialize, however, unless asset managers' MBS holdings exhibit pronounced negative convexity.

An index-level view offers a first, seemingly benign perspective on the problem, but it proves fundamentally misleading. Bloomberg provided a comprehensive overview of the December 2023 U.S. agency MBS universe[16] earlier this year. Mortgage rates and MBS yields at year-end differed by only about 10 bps from their mid-year values (where the Fed paper's analysis concludes), making this data applicable. Imported from the presentation, a graph and a table paint a reassuring picture.

[16] Jain, V. "Diving into the Mortgage Pool," (2024). Bloomberg Professional Services, Systematic Strategies.

Figure 6 illustrates that, through 2023, index duration has hovered around 6 years, the highest level in the series since 1989. More critically, convexity is also near all-time highs, with values close to zero. These are both reassuring signs. The high duration indicates that holders overweight MBS have minimal, if any, duration to hedge, as the US Agg has a similar duration profile. Near-zero convexity suggests that the gap to the convexity of duration-matched Treasuries (+0.35, at the time of writing) is small enough for the foreseeable future. Barring very large rate fluctuations, re-hedging pressures should be very manageable.

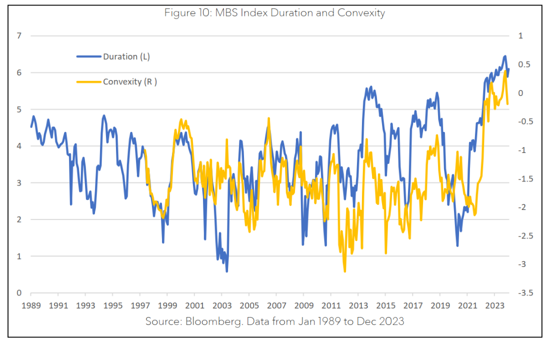

Fig. 6

Figure 8: MBS Index Analytics by Coupon*

Coupon	Weight	Yield	Duration	OAS	Z Spread	Option Cost	Convexity
1.5	2.59	4.6	7.8	53.4	67.1	13.7	0.8
2	22.90	4.5	7.5	42.8	55.4	12.7	0.6
2.5	18.74	4.6	7.1	42.0	54.7	12.7	0.6
3	12.21	4.6	6.2	46.4	58.8	12.5	0.4
3.5	9.00	4.6	5.7	43.8	60.6	16.8	0.2
4	6.91	4.7	5.4	42.3	66.6	24.3	-0.3
4.5	5.10	4.8	4.8	42.3	85.6	43.3	-0.7
5	4.79	5.1	4.1	45.9	104.8	58.9	-1.5
5.5	4.26	5.2	3.3	49.9	126.1	76.1	-2.6
6	2.92	5.3	2.7	61.7	149.0	87.3	-2.8
6.5	1.53	5.4	2.2	78.4	166.1	87.8	-2.4
7	0.25	5.6	1.8	103.0	193.8	90.8	-1.4

Source: Bloomberg. Data from 29th Dec 2023. *The table excludes the 15 year mortgages

Fig. 7

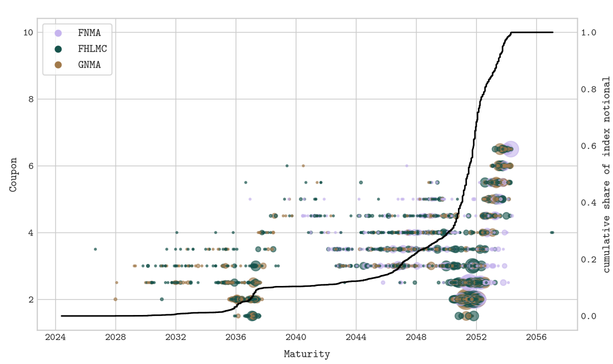

Fig. 8

The table in Figure 7 provides further insight into these two aspects by breaking down the MBS universe by coupon. Well over half of outstanding 30-year MBS have coupons of 3% or less. For these securities, the prepayment option is so far out of the money that they effectively behave as straight bonds locally, resulting in durations of 6+ years. For the same reason, they also exhibit positive convexity at current yield levels. Given their substantial weight, this has a pronounced effect on the index, driving both duration and convexity to historic highs.

A final index-level view, in Figure 8, plots the maturity and coupon of all outstanding MBS issued by one of the three agencies and with an index weight of at least 1 bp. Since nearly all MBS have a contractual maturity of either 30 years or 15 years, this plot allows for direct inferences about the issuance date of each security. The COVID-19 refinancing wave, with its distinctive cluster of coupons between 1.5 and 3, accounts for the bulk of outstanding MBS. Higher interest rates since then have slowed new housing construction and thus mortgage demand, which, together with a logical lack of voluntary early repayments, led to modest gross and net issuance since mid-2022.

To summarize, if the composition of AMs' MBS portfolios were to mirror the broader index, then risks to basis trades from re-hedging due to MBS duration shifts would be minimal, regardless of any sector over- or underweights.

* * *

Even setting aside direct access to dealer-to-client flow data, a single row comparison in Figure 1 signals that the index aggregate is not the whole story. Compare the agency MBS duration of index funds (top) to futures holders (bottom) in December 2019, June 2021, and June 2023. The duration trend for index funds naturally tracks the actual index duration in Figure 6, with increases between the two dates totaling almost 2 years. Futures holders, on the other hand, started out in 2019 with a duration 0.2 years above the index duration, which widened to 0.4 by 2021. However, between 2021 and 2023, the duration of futures holders decreased by 0.5, while index funds saw an increase of 1.3, a significant divergence.

The mechanics driving this divergence become clear when examining a single, representative fund: Capital Group's *The Bond Fund of America*. As the largest IID fund, its credentials are ideal: $75 bn in net assets, a $31 bn net long Treasury futures position, and, critically, a 40% allocation to MBS—a 15-percentage-point overweight versus the US Agg. This makes it a quintessential example of the strategy driving the basis.

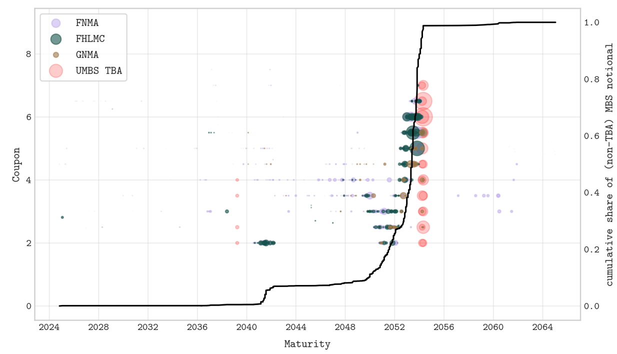

Fig. 9

As is evident, the maturity distribution deviates significantly from Figure 8. Only 25% of the value of non-TBA MBS exposures has maturities prior to 2052, with the hockey-stick effect not kicking in until mid-2052. In stark contrast to the index, a substantial 62% of the dollar value held has a 5-coupon or higher.

The explanation for this composition lies in portfolio flow dynamics, where an active MBS allocation functions essentially as an inventory queue. Rather than stemming solely from deliberate benchmark deviations,[17] the above profile emerges organically from the mechanics of portfolio turnover: newly acquired holdings naturally reflect recent, actively traded issuance, whereas seasoned bonds have already settled into HTM portfolios. Reductions in MBS allocations can ideally be achieved by stopping purchases and redeploying received coupons and prepayments elsewhere, with more rapid adjustments addressed through TBAs.

Starting with a duration similar to the index at the end of 2019, IID funds began redeploying the heavy prepayment flows of 2020 elsewhere when the OAS contracted. As OAS approached zero in H1 2021, they sharply reduced MBS exposure. While prepayment run-off drove a substantial portion of the exit, slowing refinancing speeds meant this natural cash flow had to be supplemented with targeted outright sales. This behavior had two opposing effects on duration. Selling existing high-coupon, low-duration bonds tended to lengthen the portfolio's average duration, while forgoing reinvestment in new, longer-duration issues—which were entering the index—exerted a countervailing drag. These forces largely offset, resulting in only a slight extension of IID funds' aggregate MBS duration and widening the positive gap to the index by 0.2 years.

Crucially, since IID funds entered 2022 underweight MBS, they had significant leeway to increase their exposure

[17] For an intentional reason, see Boyarchenko et al. (2014): *"Understanding Mortgage Spreads"*. Prepayment risk premiums expand primarily when mortgages trade with positive moneyness. An active shift into newer, higher-coupon issuance, particularly once peak yields subside, could plausibly position funds to earn this extra OAS.

when OAS surged—and they did, nearly doubling their agency mortgage holdings to more than 15 percentage points above the benchmark. The higher interest rate environment in 2022–2023 led to the issuance of MBS with coupons in the 4–6% range. As a result, these low-duration bonds now comprise a material portion of IID mutual funds' holdings, which explains why their aggregate MBS duration is lower.

Figure 7 confirms the characteristics of the securities these funds acquired. The 4–6% coupon range contains the 3.78-year duration reported for IID funds in mid-2023, and features an average convexity of -1.6 that can drop below -2 for higher coupons. While not a historical extreme, this profile is worlds apart from the benign picture of the index, making a substantial futures hedge unavoidable for any benchmark-adhering fund.

VII. Quantifying $\mathrm{d}D/\mathrm{d}r$

Q: How does duration change with rates?

An asset's duration is not static; its sensitivity to interest rate changes is the engine of the dynamic hedging requirement at the core of this paper's thesis. This sensitivity is captured by $\mathrm{d}D/\mathrm{d}r$, the derivative of Duration (D) with respect to rates, which emerges from the fundamental relationship between an asset's Duration itself and its Convexity (C).

$$D = -\frac{1}{P}\frac{\mathrm{d}P}{\mathrm{d}r} \qquad\qquad C = \frac{1}{P}\frac{\mathrm{d}^2P}{\mathrm{d}r^2}$$

$$\frac{\mathrm{d}D}{\mathrm{d}r} = \frac{\mathrm{d}}{\mathrm{d}r}\left(-\frac{1}{P}\frac{\mathrm{d}P}{\mathrm{d}r}\right) = \frac{1}{P^2}\left(\frac{\mathrm{d}P}{\mathrm{d}r}\right)^2 - \frac{1}{P}\frac{\mathrm{d}^2P}{\mathrm{d}r^2} = D^2 - C$$

The formula $\mathrm{d}D/\mathrm{d}r = D^2 - C$ decomposes duration sensitivity into two distinct forces.[18] The D^2 term reflects a price-base effect: as bond prices rise, the price denominator mechanically compresses percentage duration. The C term measures pure price-yield curvature. While positive convexity in conventional bonds offsets the D^2 effect to lengthen duration as yields fall, an asset with negative convexity flips the sign of C, reinforcing D^2 and accelerating duration contraction.

[18]Please note: Many commercial analytics providers (Bloomberg included) divide convexity by 100, displaying an OAC of -1.6, for example, rather than -160. That neatly simplifies bond price second-order Taylor approximations by absorbing the $(0.01)^2$ factor of a 100 bps move. Here, however, evaluating $\mathrm{d}D/\mathrm{d}r$ requires the unscaled units, meaning terminal numbers are multiplied by 100 throughout our calculations.

A stylized 10-year U.S. Treasury bond illustrates how these magnitudes balance in sovereign debt: with a Modified Duration of 8 *years* and Convexity of approximately 78 $years^2$,

$$\mathrm{d}D/\mathrm{d}r = 8^2 - 78 = \mathbf{-14}$$

A 100 bps upward rate shift shortens the bond's duration by about 0.14 years. For high-coupon MBS, however, large negative convexity dominates this equation, creating the precise hedging vulnerability this paper investigates.

The key inputs for the $\mathrm{d}D/\mathrm{d}r$ calculation are mapped in the following figures. For agency MBS, we establish a baseline using year-end 2023 Bloomberg index data, previously detailed in Figure 7. Crucially, we have verified this profile remains stable through mid-2024 using the convexity profile of UMBS TBA, the market's most liquid segment (Figure 10). The corresponding duration and convexity for U.S. Treasuries is shown in Figure 11.

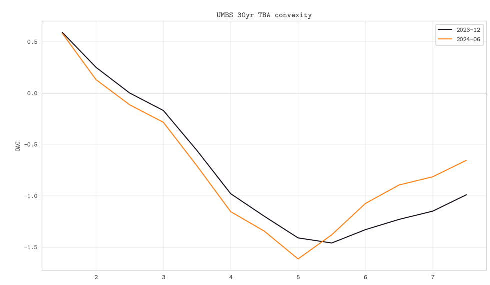

Fig. 10: UMBS TBA Convexity Profile (Current Coupon): Year-End 2023 and Mid-2024

The resulting $\mathrm{d}D/\mathrm{d}r$ metric is graphed in Figure 12 for the Treasury curve by maturity and for agency MBS by coupon. This comparison clearly demonstrates the dominance of large negative convexity in the MBS case, with $\mathrm{d}D/\mathrm{d}r$ values peaking near 300. Although long-term Treasury bonds exhibit greater absolute convexity, this effect is largely offset by the opposing and equally rising impact of D^2, keeping the Treasury curve's $\mathrm{d}D/\mathrm{d}r$ much closer to zero.

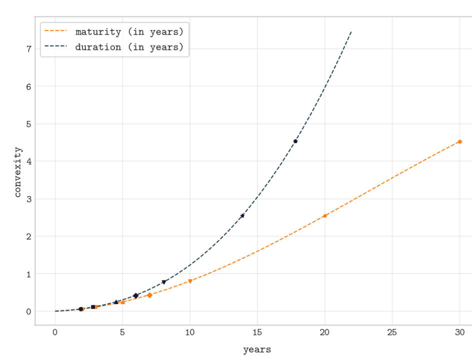

Fig. 11: U.S. Treasury Duration and Convexity Across the Curve

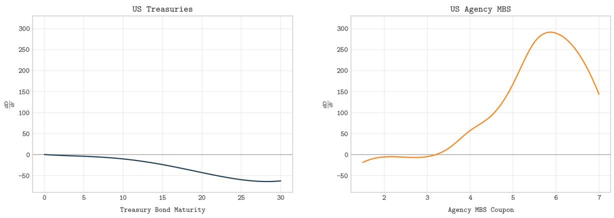

Fig. 12: dD/dr Comparison: U.S. Treasuries vs. Agency MBS

The offsetting hedge is a portfolio of Treasury futures. Table II details their duration and convexity characteristics.[19] The bottom rows aggregate these contract-level statistics, calculating the weighted risk metrics for mutual funds' actual gross and net long Treasury futures positions. For the net long positions, this results in a weighted duration of 3.04 years and a convexity of just 0.16. To ground these aggregate figures, the final column provides a concrete case study for the Bond Fund of America (ABNDX), the representative fund analyzed previously.

TABLE II: Treasury Futures Characteristics and Fund Exposures

Contract	~DUR	~CVX	MF longs	MF net	ABNDX
2y (TU/ZT)	1.8	0.05	0.45	0.61	0.38
5y (FV/ZF)	4.3	0.22	0.28	0.35	0.46
10y (TY/ZN)	7.1	0.46	0.16	0.09	0.14
10y Ultra (UXY/TN)	8.2	0.76	0.04	−0.07	−0.04
20y (TWE)	14.5	2.54	0	0	−0.05
30y (US/ZB)	14.3	2.54	0.03	0	0
Ultra Bond (UB/WN)	18.5	4.05	0.05	0.02	0.11
Weighted duration			**4.83**	**3.04**	**4.65**
Weighted convexity			**0.45**	**0.16**	**0.48**

The stark divergence in dD/dr between MBS and their hedges provides the mechanism for the risks we now quantify. The evidence shows that futures-using asset managers' MBS exposures are heavily skewed towards high-coupon securities—with significantly higher dD/dr—than the MBS index. Coupon composition aside, they are also overweight MBS as an asset class. These firms rely on long Treasury futures to cover duration shortfalls. However, since the dD/dr of futures cannot offset the far greater dD/dr of their MBS holdings, they are exposed to significant re-hedging pressures should rates decline. The magnitude of these pressures is the subject of the next section.

[19] Futures contracts use conversion factors to create a standardized measure of deliverable bonds, allowing for comparison as if they were a hypothetical 6% coupon bond. While the conversion factors themselves are crucial in determining the futures price, the duration and convexity of the futures contract are, to a first approximation and neglecting certain technicalities (CTD switches, delivery timing and accrued bond interest), equal to the duration and convexity of the cheapest-to-deliver bond.

VIII. Quantifying the Re-hedging Requirement

Q: What is the DV01 cost of the high-coupon skew?

To quantify this risk, it is necessary to determine the volume of high-coupon MBS held by asset managers who use Treasury futures to manage duration. Because this data is not publicly aggregated, we conducted a direct, security-level analysis of N-PORT filings for the entire U.S. mutual fund universe.[20] The following analysis presents these findings. Readers focused on the final estimates may prefer to skip to Table III.

Isolating the Target Cohort

The analysis is grounded in the complete universe of 12,500 N-PORT reporting funds, representing a total AuM of $33 trillion. Within this universe, we identify aggregate agency MBS holdings of $1 trillion[21] and gross UST futures notional of $640 billion ($470 billion long, $170 billion short).[22] The market is highly concentrated: while the largest 2,000 funds account for over 85% of these exposures, the net long futures positions are dominated by the top 800, with the bulk of this exposure held by just the top 100. A notable feature in Figure 13 is the near-monotonic rise of the net long futures position (green line), underscoring the systemic, rather than idiosyncratic, nature of the duration catch-up phenomenon.

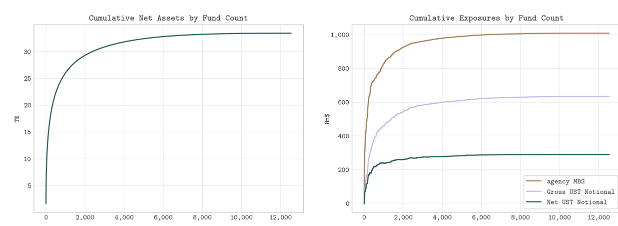

Fig. 13

[20] This bottom-up approach was necessary as existing research, including the Federal Reserve paper, does not provide this level of granularity. For context, N-PORT filings are comprehensive quarterly portfolio reports. Processing the Q1 2024 data for all 12,500 filers involved parsing over 100 GB of raw XML data—a non-trivial task given the SEC's EDGAR download policies. Our analysis extracted general fund information (`<genInfo>`, `<fundInfo>`) and position-level data for all agency MBS and Treasury futures holdings (`<invstOrSec>`).

[21] The N-PORT derived AUM of $33 trillion is consistent with the Federal Reserve's Z.1 figure of $29.6 trillion for US mutual and exchange-traded funds, once Z.1's narrower scope (ICI-reporting companies) is considered. Our larger MBS figure of $1 trillion (vs. Z.1's $708 bn) is primarily due to our inclusion of TBA positions, which are critical to capturing asset managers' full effective exposure to the mortgage market.

[22] The sizing of the futures-user cohort and the resulting risk estimates should be considered conservative. The underlying N-PORT data for futures counterparties is subject to significant reporting variance, making direct name-based filtering unreliable. Our analysis therefore uses a standardized ISO country code screen, which provides a consistent baseline but conservatively excludes positions where filers use non-standard codes for U.S. exchanges. We estimate this understates the true notional exposure by approximately 25%, meaning the final calculations in Table III represent a robust lower bound on the systemic risk.

The central task is to isolate the cohort of funds that structurally employ Treasury futures to manage duration benchmarks—the *futures users*.[23] Our definition rests on a sequence of precise, observable criteria: funds must hold non-zero gross UST futures positions, maintain a net long exposure, and, critically, exceed a minimum threshold (x) for the net futures notional relative to AUM to distinguish structural hedging from minor tactical tilts.[24]

The optimal value for this threshold is a data-driven judgement. As Figure 14 illustrates, raising x from zero eliminates funds with incidental futures usage while having a minimal impact on the aggregate exposures of the core cohort. We select a threshold of $x \geq 2.5\%$, a level that captures over 90% of the relevant MBS exposure held by net-long futures users while being sufficiently material to represent a deliberate portfolio construction choice. This filtering framework yields a final cohort of approximately 700 funds, which are the focus of the subsequent analysis.

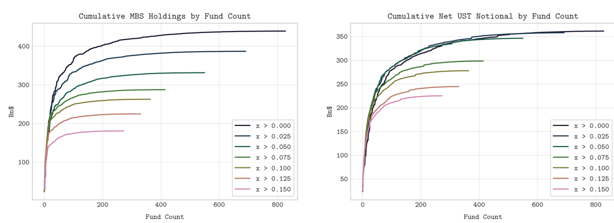

Fig. 14

Uncovering the High-Coupon Skew

The 700 funds identified as futures users hold a combined $417 billion in agency MBS.[25] The primary risk factor explored here lies not in the size of this portfolio, but in its specific composition, which reveals a stark divergence from the benchmark index.

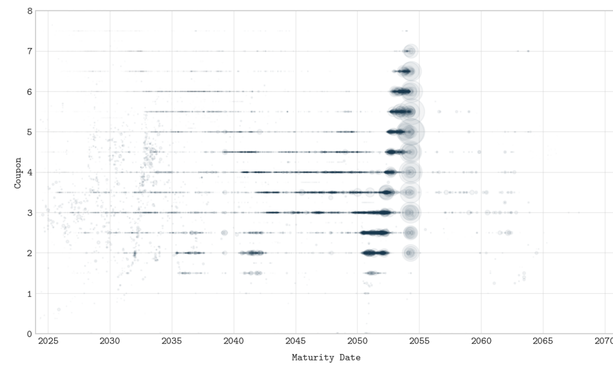

Fig. 15

The cohort's median coupon of 3.75% stands in stark contrast to the index's 2.75%. As the maturity-coupon scatter plot (Figure 15) shows, this divergence is driven by a hybrid composition: the portfolio is roughly split between low-coupon bonds from the COVID-era refinancing wave and more recent, high-coupon issuance. This positions futures users' aggregate holdings between the low-coupon bias of the overall index and the high-coupon concentration of the most active funds.

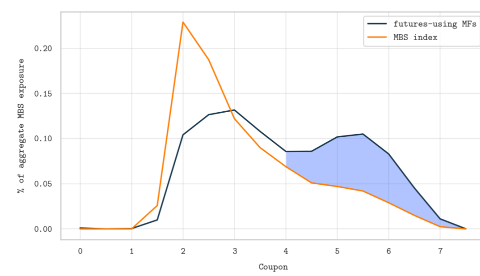

Fig. 16

To precisely quantify this high-coupon overhang, we consolidate the scatter plot into the coupon distribution shown in Figure 16. The resulting pattern is a pronounced double-humped shape, setting it distinctly apart from the MBS index. The large excess holdings in the 4%+ coupon range, shaded in the figure, collectively account for 26% of futures users' MBS exposures. In other words, over a quarter of their MBS portfolio consists of high-coupon, negatively convex securities that are materially underrepresented in the benchmark.

This compositional skew has a dramatic effect on the portfolio's duration sensitivity, $\mathrm{d}D/\mathrm{d}r$.[26] Summing the coupon-weighted contributions, as illustrated in Figure 17, yields a $\mathrm{d}D/\mathrm{d}r$ of 94 for futures-using portfolios, compared to just 38 for the MBS index. For a 100 bps rate decline,

[23] Ideally, one would identify futures users through direct inquiry or a fund-by-fund high-frequency time-series analysis of whether cash-only duration deviations are reliably closed via futures versus a benchmark. Given the impracticality of these approaches, our cross-sectional methodology is designed as the most robust alternative. The filtering logic is sequential: we first require a non-zero gross futures position to eliminate nearly all equity funds and fixed-income index funds. We then require a net long position, as this aligns with the objective of offsetting the shorter-duration profile of spread assets like MBS relative to benchmarks such as the US Agg.

[24] The purpose of the threshold x is to solve a signal-to-noise problem: separating funds using futures for structural, benchmark-driven hedging from those using them for minor, tactical duration timing. Traditional fixed-income funds rarely take large tactical duration views, as performance is typically measured against a benchmark or peer group. Setting a non-trivial threshold therefore effectively isolates the target population. While no single threshold is uniquely definitive, the 2.5% level represents a reasoned choice that is large enough to indicate intentionality and warrant CIO-level attention, yet low enough to not exclude significant portions of the structurally hedging universe.

[25] This composite exposure was compiled from approximately 100,000 line items across the relevant N-PORT filings, which consolidated into over 40,000 unique CUSIPs. The 400 largest positions, including TBAs, account for roughly half of the total exposure.

[26] A parallel shift in U.S. agency yields is assumed. A more granular analysis could decompose this into agency spreads and risk-free yields, but a decline in Treasury yields is the most likely catalyst for the rapid re-hedging scenario under consideration.

this means the duration of futures-users' MBS holdings shortens by an additional 0.56 years—a sensitivity nearly three times that of the benchmark.

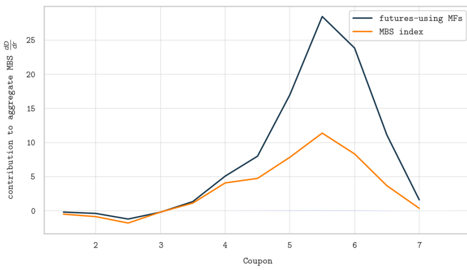

Fig. 17: Coupon-weighted contributions to mutual funds' overall $\mathrm{d}D/\mathrm{d}r$.

Amplifying Duration Sensitivity: The Overweight Factor

The analysis thus far has focused on the **compositional** $\mathrm{d}D/\mathrm{d}r$ gap between futures-users' portfolios and the MBS index.[27] This comparison, however, underestimates the total re-hedging risk. The full sensitivity arises from two distinct sources: the first is the compositional skew towards high-coupon bonds, and the second, which we address here, is their significant **overweight** to the agency MBS asset class as a whole.

The key to this amplification lies in the benchmark assets being displaced. When futures users are overweight agency MBS, they are necessarily underweight other benchmark components—primarily shorter-dated Treasury bonds.[28] As these Treasuries have negligible convexity, their $\mathrm{d}D/\mathrm{d}r$ is effectively zero. Therefore, for the portion of the MBS portfolio that constitutes an overweight, the relevant $\mathrm{d}D/\mathrm{d}r$ gap is not relative to the MBS index, but to a baseline of zero. This portion contributes its entire $\mathrm{d}D/\mathrm{d}r$ of 94 to the total gap.

To quantify this, we establish a baseline proportion for the overweight. Position data confirms that futures users—particularly within the core IID fund universe—maintain a substantial overweight relative to their benchmarks.[29] On this basis, **two-thirds** of their MBS holdings are treated as the index-replicating portion, with the remaining **one-third** reflecting the structural overweight.

This leads to our final estimate for the total $\mathrm{d}D/\mathrm{d}r$ gap, calculated as a weighted average of the two components:

- **Compositional Portion (2/3 of holdings):** Contributes a $\mathrm{d}D/\mathrm{d}r$ gap of **56**.
- **Overweight Portion (1/3 of holdings):** Contributes a $\mathrm{d}D/\mathrm{d}r$ gap of **94**.

The resulting total estimated excess $\mathrm{d}D/\mathrm{d}r$ is 69, a figure more than 20% higher than the compositional analysis alone would suggest.

The Price of Convexity: DV01 Shortfall and Notional

An excess dD/dr of 69 quantifies the duration shortfall a portfolio will experience when interest rates fall. The most practical way to assess this re-hedging pressure is to translate it into DV01 (dollar duration per basis point).

The unit calculation is straightforward. For a representative \$1 billion portfolio, a 100 bps parallel decline in rates generates a DV01 shortfall of:

$$\frac{69 \times 0.01}{10,000} \times \$1 \text{ bn} = \$69,000$$

Scaling this to the futures users' actual \$417 billion MBS portfolio amplifies the DV01 shortfall to **\$29 million**. To put this figure in perspective, data from the Federal Reserve shows their entire existing net long Treasury futures position (∼\$400 billion) carries a total DV01 of only \$120 million. **A 100 bps rally would therefore generate a duration gap equivalent to nearly a quarter of their entire current hedge.**

To convert this DV01 shortfall back into a required notional hedge, we must determine the weighted-average duration of the futures basket these managers employ. Our analysis of the N-PORT data for the most concentrated futures users finds a weighted duration of 4.15 years.[30] This implies a DV01 of approximately \$415,000 per \$1 billion of futures notional. At this conversion factor, neutralizing the \$29 million DV01 shortfall would require an additional **\$69 billion** in net long Treasury futures notional from mutual funds alone.

[27] Evaluating the gap relative to the MBS index is necessary because benchmark duration also contracts as rates fall, an effect that requires no offsetting re-hedging by the asset manager.

[28] As shown in Figure 1 of this document and Table 5 of the Federal Reserve's working paper, these shorter-dated Treasuries exhibit negligible convexity.

[29] Portfolio disclosures for the largest IID funds in mid-2024 show an average agency MBS allocation of 38%, compared to the US Agg's 26%.

[30] This figure is derived from position-level N-PORT filings for the most concentrated futures users, defined as funds with net Treasury futures notional/AUM of at least 5% and agency MBS exposures exceeding 30% of net assets. The notional amounts by ticker for this cohort are illustrated in Figure 18. This approach provides a more precise, data-driven measure than the broader sector averages discussed in Table II.

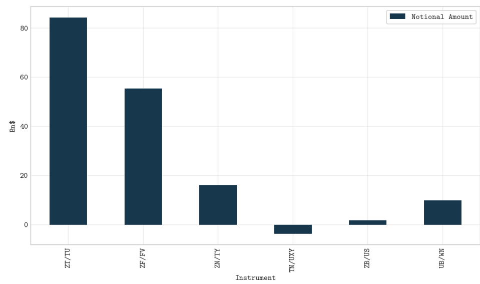

Fig. 18: Notional amounts by ticker for the most certain futures users.

Sizing the Opaque: The Separate Account Triangulation

To construct a complete picture of the systemic risk, our mutual fund findings must be scaled to account for the opaque separate account universe.[31] This presents a significant analytical challenge due to the lack of public data. We conducted a rigorous triangulation of top-down AUM data and bottom-up market signals to overcome this opacity and establish a robust estimate.[32] Our analysis concludes that futures-using separate accounts are of approximately equal size to their mutual fund counterparts, yielding a total scaling factor of 2.0x for the asset manager universe.

The Bottom Line: Sizing and Validating the Aggregate Risk

The final step is to synthesize these findings. Applying the 2.0x scaling factor to the mutual fund cohort's DV01 shortfall ($29 million) yields a total requirement of $58 million for the entire asset manager universe. To neutralize this duration gap following a 100 bps rate decline, asset managers would need to acquire an additional **$139 billion** in net long Treasury futures notional.

Table III consolidates our estimates of asset managers' hedging pressure in and around the high coupon area. With falling interest rates, the sensitivity of futures users' hedging positions, measured by $\mathrm{d}D/\mathrm{d}r$, will change. Although we will not quantify this effect here, intuitively, $\mathrm{d}D/\mathrm{d}r$ changes relate to how managers' short high-coupon MBS prepayment options move into the money, shedding their gamma, i.e., MBS convexity pulls to zero. Visually, this shift is represented by the hump in the right panel of Figure 12 moving westward, leaving the high-coupon area behind and impacting coupons that futures users are no longer overweight in (see Figure 16). In practice, our $\mathrm{d}D/\mathrm{d}r$ estimate remains a solid approximation for yield declines of up to 130 bps from mid-2024 levels, beyond which it tails off.

As a sanity check, a more direct calculation confirms this conclusion. Futures-using AMs hold a substantial amount of excess high-coupon (4s–7s) MBS, totalling around $260 bn. Since they are short the corresponding prepayment options, they will lose much of the duration of these securities as interest rates decline. Specifically, if rates were to drop sharply, the duration of these high-coupon assets would decrease from a mean of 3.3 years currently to ∼0.5 years until they are refinanced. To offset the 2.8 years in lost duration, AMs would need to acquire an additional $180 bn of Treasury futures notional, using the same 4.15 year duration Treasury futures basket as earlier. This estimate is consistent with the $139 bn figure for the first 100 bps decline in rates.

* * *

The aggregate figures, while substantial, are not dispositive. The critical question remains: is this a genuine systemic risk, or will it dissipate as uncoordinated, idiosyncratic noise across hundreds of individual funds?

To answer this, we must move from the macro to the micro. We return to the IID fund universe for context, where 38% of net assets were allocated to agency MBS in mid-2024, and the total $\mathrm{d}D/\mathrm{d}r$ gap stood at 69. A 100 bps interest rate decline accordingly leads to a 0.26-year reduction in duration for the average IID fund. The likelihood of re-hedging quickly depends on how this 0.26-year change measures up against typical duration fluctuations around a benchmark. To determine the typical scale of such fluctuations, we sample the 10 largest IID funds and calculate quarterly shifts in their relative modified duration compared to the Agg, beginning with the introduction of N-PORT at the end of 2019.

[31] See the January 2024 TBAC presentation (p. 12), which notes that based on eVestment data, separate accounts and mutual funds collectively account for nearly 90% of AUM for a cohort of the industry's largest firms, underscoring their systemic relevance.

[32] The estimate of approximate parity (1.0x) is derived by triangulating two conflicting signals. The first is a top-down AUM sizing anchored to the consultant-advised institutional universe, which accounts for an estimated 62% of the total separate account market. By using this core, unscaled segment as our basis, the analysis establishes a conservative lower bound for the AUM sizing, which suggests separate accounts are the larger cohort (approx. 1.3x mutual funds). This sizing begins with the total Assets Under Advisement from the Pensions & Investments survey ($47.9 trillion), which is then sequentially filtered for liquid assets (assuming a 20% alternatives allocation), U.S. client assets (64%), and finally U.S. fixed income mandates (24%, per Gerakos et al.), yielding an estimated universe of $5.8 trillion. This figure establishes a baseline of near-parity when set against the relevant active mutual fund universe of ∼$5.5 trillion; the final ∼1.3x factor is a composite that adjusts for the significantly higher prevalence of intermediate-core and core-plus mandates within the institutional space—a dynamic independently corroborated by data from S&P's Institutional SPIVA Scorecard.

This top-down signal is contrasted by a bottom-up, market-based signal from 2024 CFTC futures data. In the 2- and 5-year contracts most relevant for this hedging activity, mutual funds account for a combined 59% of total net long positions, implying separate accounts are the smaller cohort in this specific context (approx. 0.7x). The final estimate of approximate parity (1.0x) is an equal weighting of these two analyses.

TABLE III: Quantifying AM Re-hedging Pressure from High-Coupon MBS

Description	Value
MBS Characteristics	
MBS High Coupon Area (HCA)	4s–7s
Futures-users' excess holdings in HCA (as % of total MBS)	26 %
Duration and Composition Analysis	
Futures-users' MBS dD/dr gap vs. MBS index (coupon composition only)	56
Overweight MBS Holdings (as fraction of total)	1/3
MF futures-users' total dD/dr gap (coupon composition & excess holdings)	69
Financial Metrics & Scaled Risk	
MF futures-users' MBS exposures (Q1-2024)	$417 bn
MFs' DV01 Shortfall (vs. benchmark, -100 bps move in HCA)	$28.8 mn
Size of futures-using separate accounts relative to MFs	100 %
AMs' total DV01 shortfall (MFs × 2.0)	$57.6 mn
Required Treasury Notional Hedge (-100 bps move in HCA)	
Weighted Duration of Treasury Futures Hedge	4.15 yrs
Required Notional (Mutual Funds)	$69.4 bn
Required Notional (All AMs)	$138.8 bn

MBS: Mortgage-Backed Securities; HCA: High Coupon Area; MF: Mutual Fund; AM: Asset Manager
dD/dr: Change in duration with respect to change in interest rate
DV01: Dollar Value of 1 basis point change

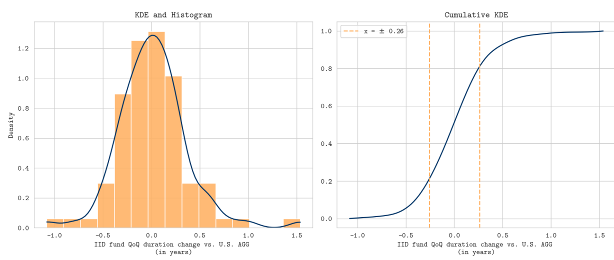

Fig. 19

Fig. 19 displays the result. Changes are, as expected, zero-centered, with quarterly variations of more than half a year being rare. The right panel includes a CDF approximation, showing that an absolute change of 0.26 years exceeds 60% of the absolute quarterly changes observed over the past five years.[33]

For IID funds, an MBS-driven duration shortfall of this magnitude is large enough to compel active re-hedging via long Treasury futures. This dynamic is reinforced by the speed of rate moves during stress episodes.[34] Even if concentration varies across individual managers, the aggregate size of the intermediate fund sector ensures these micro-level pressures remain systemically material. Consequently, a sharp decline in yields is poised to trigger substantial re-hedging flows.

[33] At a quarterly frequency, most significant deviations in fund duration reflect active views. One does not inadvertently under- or overshoot index duration by 0.3 years in a quarter. That is, as long as the duration shortening due to agency MBS holdings does not fortuitously coincide with an intended duration change, it will very likely be offset.

[34] The comparison in Fig. 19 is to the net duration changes over a full quarter. In a crisis scenario, however, rates can move sharply in a matter of days. A sudden, mechanical duration shock of this magnitude would be far more significant relative to normal, deliberate fund activity over such a short period, making an immediate re-hedging response more likely.

IX. Impact Analysis: Stock, Flow & The March 2020 Precedent

Q: So, what's the impact?

Assessing the potential impacts of AMs' re-hedging pressures is challenging due to their highly contingent nature. Rather than pursuing a single, definitive impact figure, a more productive approach returns to first principles, examining the underlying dynamics and incorporating as much historical context as possible.

The Stock Perspective

A charitable view of potential risks would compare the additional hedging demand to the current outstanding size of AMs' long and hedge funds' corresponding short UST futures positions. These, in the vicinity of $1 tn for both, substantially outweigh AMs' estimated additional hedging demands of $139 bn for the first 100 bps of rate declines, and around $180 bn in a worst-case scenario. Although not insignificant, a 10–20% increase on top of existing positions is unlikely to pose a systemic risk on its own.

Absorbing an additional $200 bn at already elevated basis trade volumes would certainly test the capacity of current arbitrageurs and bank repo lenders more than a comparable increase from a lower baseline. However, the growth, new launches, and spinouts of several strong-lineage FIRV funds in recent years mitigate this concern. With solid historical track records, these funds are well-positioned to raise and deploy additional capital into a slowly widening basis scenario.

Recent history provides a useful benchmark for the market's absorption capacity.[35] Over the past 2.5 years, AMs' net long positions have increased by approximately \$1 tn—an average of ∼\$1.6 bn per business day. Peak daily flows were, of course, higher, likely in the high single-digit billions. The relatively stable economic backdrop during this period has allowed these demands to be accommodated without significant disruption, although the cash-futures basis has fluctuated.

This benign conclusion, however, rests on two critical—and often fragile—assumptions: gradualism and market stability. Stability, in this context, implies a functioning arbitrage channel, with basis traders and their bank counterparties free from acute funding or balance sheet pressures. The entire analysis changes when these assumptions break down, and flows are compressed into a period of market stress.

The Flow Perspective

The stock-level comparison breaks down at the margin. In a compressed timeframe, market clearing is governed entirely by flow, meaning the static inventory of existing positions ('who already owns what') is largely secondary to the immediate price impact of a sudden re-hedging demand.

The flow perspective centers on precise questions of market impact. What price effect on the cash-futures basis is generated by an \$$x$ bn:

- one-off extra net demand for Treasury futures by AMs?
- one-time sale of cash Treasuries by some unrelated, cash-starved end-holder?
- one-time unwinding of a basis trade by an RV hedge fund?

And:

- How do these effects differ under various market conditions, beyond average impacts?
- What happens if the \$$x$ bn transaction is repeated over several days, rather than being a one-off event?

Here, unlike in liquid equity markets, modeling market impact from first principles is notoriously difficult. Fixed income relative value lacks the two structural pillars that make equity impact models tractable: (1) vast, continuous, high-frequency datasets across a broad cross-section, and (2) a deeply fragmented ecosystem where liquidity on the short- to mid-frequency end of the alpha curve is provided by a diverse pool of automated market participants.

On the empirical side, substantial hedge fund basis trade volumes have occurred across only two distinct windows—a brief period in the late 2010s and the post-2022 cycle—yielding barely half a decade of data, nearly all of it during economic expansions without severe financial shocks. High-frequency tick data is similarly scarce. On the structural side, while equity impact functions as an aggregation game across broad liquidity pools, Treasury relative value is governed by a small, concentrated cohort of roughly 40 discretionary desks,[36] an even smaller group than the asset managers driving demand. Consequently, liquidity absorption cannot be modeled as a continuous statistical process,[37] limiting the direct transferability of standard microstructure heuristics, such as the square-root law or conventional TCA frameworks.

* * *

Given these analytical constraints, a purely quantitative, top-down model of market impact is unlikely to be robust. A more fruitful approach is to build a qualitative, bottom-up understanding by examining the decision-making framework and capacity constraints of the arbitrageurs themselves.

Basis books are typically centralized within an organization under close CIO involvement for several reasons: (1) their balance sheet-heavy and counterparty-intensive nature, (2) attractive and largely uncorrelated returns, which, however, come with known yet difficult-to-hedge downside risks that make the centralized version often preferable to many pods, and (3) the difficulty of liquidating positions during periods of stress, typically impossible without incurring major losses, also calls for stronger firm backing than easier to wind down strategies.

Against this background, there is usually a predefined exposure band within which a firm's current engagement is envisioned, and within which trades can be executed without additional clearance. This band, which may leave room for a few extra hundred million or single-digit billion amounts depending on the portfolio manager, allows them to step into common dislocations during the day, provided these actions align with their overall portfolio conception.

External institutions' demands, whether for future buying or cash bond selling, that fall within this capacity are typically accommodated with minimal market impact.

[35] While applying a dynamic timeframe to a static view seems counterintuitive, it provides an essential, real-world calibration of market capacity for persistent, one-sided flows under benign conditions.

[36] Broad market aggregation produces continuous, relatively stationary impact functions across regimes, a structural feature largely absent in the idiosyncratic and event-driven FIRV space.

[37] This is not to suggest that no empirical insights can be drawn from March 2018, September 2019, March 2020, or September 2022.

However, when external flows press against these internal limits, expanding capacity typically requires clearance from senior risk management and the CIO to evaluate whether market conditions justify additional balance-sheet allocation. The resulting delay can lead to more erratic market behavior, where dislocations persist for longer periods before being corrected.

Finally, there is a hypothetical level at which the ability of RV traders to absorb basis-widening flows switches from positive to negative. This can occur due to a combination of factors, including realized PnL stress, funding, margin, and haircut pressures on either leg, as well as PM-specific or firm-wide risk reduction efforts, all conspiring to put on the block otherwise positive-EV basis packages. At this, once-in-a-decade point, market impacts escalate sharply, with the perceived arbitrageurs tapped out.

This qualitative framework provides the necessary context to assess the central question: how do the potential re-hedging pressures from AMs compare to the realized, market-moving flows observed during the March 2020 stress event?

The Lessons of March 2020

Two facts about the events of early 2020 are widely recognized:

- Basis trades incurred significant losses in March as the basis surged to around 120 bps at its peak, depending on tenor and price source.
- Hedge funds unwound a substantial portion of their sizable basis positions, while a range of other institutions sold cash Treasuries heavily for precautionary liquidity, 'dash-for-cash' reasons.

The headline figure from Table IV—over half a trillion dollars in sales for the full month—is indeed substantial. The conventional analysis—often left implicit in other accounts—infers that this aggregate sum was the direct driver of the market dislocation. On this basis, the $200 bn re-hedging risk identified in this paper would appear relatively modest, amounting to less than half the March 2020 pressure.

TABLE IV: Net Activity in Treasury Securities by Investor Type

Investor Type	**March 2020**	**March 2-13**
HF basis traders	−$148 bn	−$54 bn
Other hedge funds	−$25 bn	–
Foreign officials	−$210 bn	−$105 bn
US open-ended mutual funds	−$159 bn	−$46 bn
Total	**−$542 bn**	**−$205 bn**

This conclusion, however, collapses under a closer examination of the event's timeline. Two facts are critical:

- The cash-futures basis did not rise all month. Tenor and price source again create a bit of ambiguity, but as a chart from a Lorie Logan presentation[38] illustrates (Figure 20 for the representative 5 year tenor), the big picture is clear. Pressures began to mount entering March, peaking between 100–125 bps on the 12th or 13th, to end the month essentially where it started, near zero.

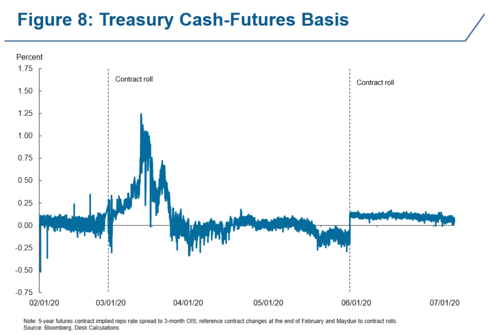

Fig. 20

- The Federal Reserve bought over $900 bn worth of Treasuries *after* March 13th. The clear implication is that a large portion of the $542 bn in sales from these key investor groups occurred in the second half of the month, directly into the Fed's bid.[39]

Isolating the critical period through March 13th reveals the true scale of the pressure.[40] The third column of Table IV shows sales of just $205 bn during this window, less than 40% of the month's total. To calculate the full market impact, the $54 bn in cash sales from basis traders must be doubled to account for the symmetrical pressure from unwinding both legs of the trade.[41] This yields the key figure for our analysis: not the misleading half-trillion monthly total, but a concentrated, acute pressure of $259 bn from these investor groups collectively—a flow that either drove, or was a primary contributor to, the

[38] Logan, L. K. "The Federal Reserve's market functioning purchases: from supporting to sustaining," (2020). Remarks at SIFMA Webinar. Federal Reserve Bank of New York.

[39] Dealer banks stepped in somewhat in the first two weeks of March, taking over some basis positions. It is difficult to put an exact number on this; likely somewhere between $25–60 bn.

[40] These are our internal daily estimates, which are post-adjusted to correctly align with lower frequency but higher accuracy statistics.

[41] Unwinding a cash-futures basis trade exerts symmetrical pressure on the basis itself. The cash sale ($54 bn) pushes the bond price down, while the simultaneous purchase to close the short futures position (an equivalent notional amount) pushes the futures price up. Both actions mechanically widen the basis, hence the doubling of the cash-leg flow to capture the total impact.

dislocation during the peak stress window.[42] This is the relevant precedent.

The re-framed data shows the basis crisis was driven in roughly equal parts by foreign official sales and the endogenous, and otherwise sensible and forward-looking, de-risking of hedge funds. Against this corrected $259 bn precedent, the potential $200 bn duration re-hedging demand from asset managers emerges as a systemic risk of directly comparable magnitude—a far more serious threat than suggested by a superficial comparison to the headline half-trillion-dollar figure.

X. What is the bottom line?

While regulatory scrutiny has rightly focused on repo financing and cash-side liquidations as the primary vulnerabilities of the basis trade, an equivalent destabilizing shock can originate from the futures leg itself: the mandatory duration re-hedging of institutional asset managers. Sized through a bottom-up analysis of mutual fund N-PORT filings and scaled across the institutional asset management sector, this requirement is propelled by a potent combination of structural overweights in agency MBS and an unprecedented concentration in recent, high-coupon vintages with deeply negative convexity.

At current interest rates, the resulting duration contraction creates an incremental hedging demand of approximately $1.4 bn in Treasury futures notional per basis point of yield decline. That translates to a $139 bn futures demand shock for an initial 100 bps rally, capping out near $180–$200 bn before option gamma decays.

To understand the systemic weight of a $200 bn flow, it must be measured against the proper historical yardstick: it approaches the concentrated $259 bn in basis-widening pressures that brought the Treasury market to a standstill in the first two weeks of March 2020. That comparison is not a prediction of a repeat crisis, but an objective benchmark of the market's empirical absorption limits.

The systemic danger lies in the programmatic and additive nature of the flow. Unlike discretionary positioning, duration catch-up is an institutional imperative enforced by benchmark tracking. In an acute market shock, forced futures purchases will hit the market precisely when traditional dash-for-cash sales are already straining dealer capacity. With the cash-futures basis trade now exceeding $750 bn—having grown by more than 50% since early 2020—systemic leverage has expanded at the exact moment that higher starting yields make a sharp, convexity-fueled rate rally entirely plausible.

[42] These considerations ignore the potential price effects of intra-sector transactions. That is, if one hedge fund seeks to delever and quickly liquidates positions while a better-capitalized hedge fund buys, the net effect is still a wider basis, with no net transactions by hedge funds as a whole. Anecdotal evidence suggests that while such effects may not have been zero, few hedge funds were size buying in March.

XI. Miscellaneous Afterthoughts

- **Historical Context (Perli & Sack, 2003):** In related literature, Perli & Sack examined the extent to which the hedging of duration shifts in agency MBS amplified swap rate movements.[43] At the peak of the GSEs' footprint, mortgage hedging flows were considerably larger than those discussed here. Crucially, however, the effects analyzed related to a linear rates product, not a basis.

  More fundamentally, the structural posture of the hedger governs where the secondary shock lands. Pre-GFC, duration-neutral MBS holders hedged by paying fixed via swaps; falling rates forced them to **reduce** this short derivative position. Today's benchmark-tracking managers, already long futures, must **add** to their longs. While both actions generate the same first-order pressure by buying duration into a rally, the relative-value impact flips: pre-GFC hedging compressed swap spreads, whereas today's programmatic re-hedging forces open the cash-futures basis.

- **Historical Context (Mid-2000s Basis):** In terms of pricing, the Treasury basis situation in the mid-2000s was the opposite of today's. While most MBS hedging was done through paying swaps, some may have been done through short Treasury futures. On the cash side, FX reserve managers were heavy bidders for Treasuries. Together, these forces resulted in the basis trading in negative territory for much of the pre-GFC period.

- **Recent Precedent (March 2020):** Did convexity-related re-hedging play a role in March 2020? Its role was marginal at best; the situation today is starkly different. The overweight of agency MBS among asset managers is more than twice as pronounced now as it was at the end of 2019. Back then, funds tended to hold slightly lower-coupon securities than the index, mitigating their relative convexity. Furthermore, the orderly yield decline throughout 2019 had already shortened MBS duration more than the sharper, but smaller, move in Q1 2020 did.

- **Market Mechanics (Futures vs. Cash):** Is an incremental basis-widening flow in futures as consequential as one in cash bonds? The greater liquidity of futures might suggest not. However, the pressure is more systemic. Whereas cash bond sales may, at first, only impact holders of that specific issue, a move in futures affects all basis traders simultaneously.[44]

- **Market Mechanics (A Stabilizing Element?):** There is a potential stabilizing element for basis trades that should accompany severe economic turmoil: a broad retreat from spread assets by asset managers. In such a flight-to-quality, they would close long futures positions and buy cash Treasuries, supporting the basis on both legs. Based on N-PORT data, however, this provided minimal relief in March 2020. Given the greater overweight in agency MBS today compared to other, more downgrade-prone spread assets, such a stabilizing flow may be even less likely now.

- **Market Structure (The Swing Voters):** A persistent non-zero basis is an expression of a supply and demand disequilibrium. Here: should mutual funds, or asset managers more broadly, be the 'swing voters' of the MBS market? And is that window already closing, with banks returning, OAS compressing from Q4-2023 peaks, and mutual fund buying recently turning flat for the first time in two years?

- **Risk Philosophy:** The analyzed case illustrates how an ostensibly pure RV trade can pick up unexpected directional exposure in its tails. Properly anticipating and managing these dynamics is what separates serious desks from the rest.

- **An Untested Point of Failure:** If systemic stress originates within Treasury futures rather than the cash market, can the Federal Reserve intervene effectively—and would it do so directly on the exchange? A forced deleveraging cascade on the CME would largely bypass central bank crisis facilities designed for primary dealers and repo markets. The operational gap between futures clearinghouses and the Fed's balance sheet is an untested point of failure. Designing a credible ex-ante backstop faces deep statutory and jurisdictional hurdles, making even a multi-year policy effort an uncertain path at best.

- **A Structural Solution:** Improvised central-bank backstops, on the other hand, can only ever be reactive; a permanent solution should aim to eliminate the vulnerability by addressing the structural demand for synthetic duration at its source. To that end, the operational blueprint for a new structural architecture is currently undergoing consultation across policy, academic, sovereign debt, and market infrastructure circles, with a dedicated paper detailing its design in preparation.

[43] Perli, R., & Sack, B. "Does Mortgage Hedging Amplify Movements in Long-term Interest Rates?," (2003). Finance and Economics Discussion Series. Washington, D.C.: Board of Governors of the Federal Reserve System.

[44] This systemic impact is amplified because the futures leg of the trade is standardized, unlike the cash leg where holdings can be distributed across various deliverable bonds, not just the cheapest-to-deliver (CTD).